

Course: B. Tech.

Branch: Civil

Semester: VII

Subject Code & Name: BTCVC701 & Design of Reinforced & Prestressed Concrete Structures

Max Marks: 60

Date: 01/ 07/ 2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.
5. IS 456:2000 is allowed to be used during design.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What are the assumptions made in the theory of torsion?	CO1	6
B) A rectangular reinforced concrete beam of size 250mm x 500mm is reinforced with 4 bars of 20mm diameter distributed at the corners with a cover of 50mm depthwise and 25mm widthwise. Further 8mm diameter 2 legged stirrups are provided at 100mm center to center. Find the torsional strength of the section. Fe415 HYSD bars are used and the transverse shear is 80kN. Concrete grade M20. Length of beam is 6 m.	CO1	6
C) What is compatibility torsion? Give an example. Define Equilibrium Torsion (or) Primary torsion?	CO1	6
Q.2 Solve Any Two of the following.		12
A) Define Short column and long column. State the criteria of minimum eccentricity to considered for column design?	CO2	6
B) Design a short R.C.C column to carry an axial load of 1600kN. It is 4m long, effectively held in position and restrained against rotation at both ends. Use M20 concrete and Fe415 steel.	CO2	6
C) Design the reinforcement in a circular column of diameter 400mm with helical reinforcement of 8mm diameter to support a factored load of 1200kN. The column has an unsupported length of 3m and is braced against side sway. Adopt M20 grade of concrete and Fe415 steel.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Explain the advantages of prestress concrete.	CO3	6
B) Elaborate Fressinet system of post tensioning.	CO3	6
C) Elaborate Magnel-Blaton system of post tensioning.	CO3	6

Q.4 Solve Any Two of the following.		12
A) Explain the various losses of prestress.	CO4	6
B) A prestressed concrete beam 250mm wide and 350 mm deep is prestressed by 12 wires of 6mm diameter located at an eccentricity of 40 mm and carrying a initial stress of 1500 N/mm^2 . The span of the beam is 8m. Calculate the percentage of losses in wires if it is pretensioned, $E_s = 210 \text{ KPa}$ and $E_c = 35 \text{ Kpa}$, relaxation of steel stress = 5% of the initial stress, total shrinkage strain is 200×10^{-6} .	CO4	6
C) A rectangular concrete beam 250 mm wide and 550 mm deep is prestressed by means of 4 numbers 12 mm diameter high tensile bars located at 200 mm from the soffit of the beam. If the effective stress in the wire is 700 N/mm^2 , what is the maximum bending moment that can be applied at the soffit of the beam for no tension to be developed in the section.	CO4	6
Q. 5 Solve Any Two of the following.		12
A) Explain the purpose of end blocks in prestressed concrete structures	CO4	6
B) Discuss the necessity of conducting structural audits for load-bearing wall type buildings.	CO4	6
C) Explain the consequences of neglecting structural audits in aging buildings. Provide examples of potential risks.	CO4	6

*****End*****